PS#12 - Taylor polynomials - Answer key

1. (AC 8a.2 Exercise 5) Suppose we know the following information about a function f:

$$f(0)=2, f'(0)=-3, f''(0)=-1, f'''(0)=0, f^{(4)}(0)=-3.$$

   (a) Determine $T_4(x)$, the degree 4 Taylor polynomial of f that is centered at $a=0$.

$$T_4(x)=2-3x-\frac{1}{2}x^2-\frac{1}{8}x^4$$

   (b) Use $T_4(x)$ to estimate $f(0.5)$. $f(0.5)\approx T_4(0.5)=0.3671875.$

   (c) State each of the Taylor polynomials $T_3(x)$, $T_2(x)$, and $T_1(x)$.

$$T_4(x)=2-3x-\frac{1}{2}x^2-\frac{1}{8}x^4$$
$$T_3(x)=2-3x-\frac{1}{2}x^2(+0x^3)$$
$$T_2(x)=2-3x-\frac{1}{2}x^2$$
$$T_1(x)=2-3x$$

2. (AC 8a.2 Exercise 7)

   (a) Determine the degree 6 Taylor polynomial approximation of $\sin(x)$ centered at $a=\frac{\pi}{2}$.
   $\sin(x)\approx 1-\frac{1}{2!}\left(x-\frac{\pi}{2}\right)^2+\frac{1}{4!}\left(x-\frac{\pi}{2}\right)^4-\frac{1}{6!}\left(x-\frac{\pi}{2}\right)^6.$

   (b) How is your result in (a) similar to the degree 6 Taylor polynomial of $\cos(x)$ that is centered at $a=0$?
   Well hey, it's totally the same, except all of the x's have been replaced by $\left(x-\frac{\pi}{2}\right)$'s.

   (c) Recall the trigonometric identity that states $\sin(x)=\cos(x-\frac{\pi}{2})$. How does this identity help you explain what you found in (a) and (b)?
   This is actually a perfect description of what has happened: we took the thing we got for $\cos(x)$ and just replaced x by $(x-\frac{\pi}{2})$, and according to this trigonometric identity, that gives us the values of $\sin(x)$.

3. (a) (AC 8a.2 Exercise 8) Note that $f(0.5)=\ln(1.5)$. Use $T_5(x)$ to estimate $\ln(1.5)$.
   $\ln(1.5)\approx T_5(0.5)=.40729166666\ldots.$

   (b) Observe that $g(1.5)=\ln(1.5)$. Use $P_5(x)$ to estimate $\ln(1.5)$.
   $\ln(1.5)\approx P_5(1.5)=.40729166666\ldots.$

   (c) Are the estimates of $\ln(1.5)$ generated by $T_5(x)$ and $P_5(x)$ the same or different? Why do you think this happens?
   They're the same! Since $g(x)$ and $f(x)$ are just horizontal shifts of each other, their Taylor polynomials should agree as long as we compensate for the horizontal shift by recentering the approximation.